

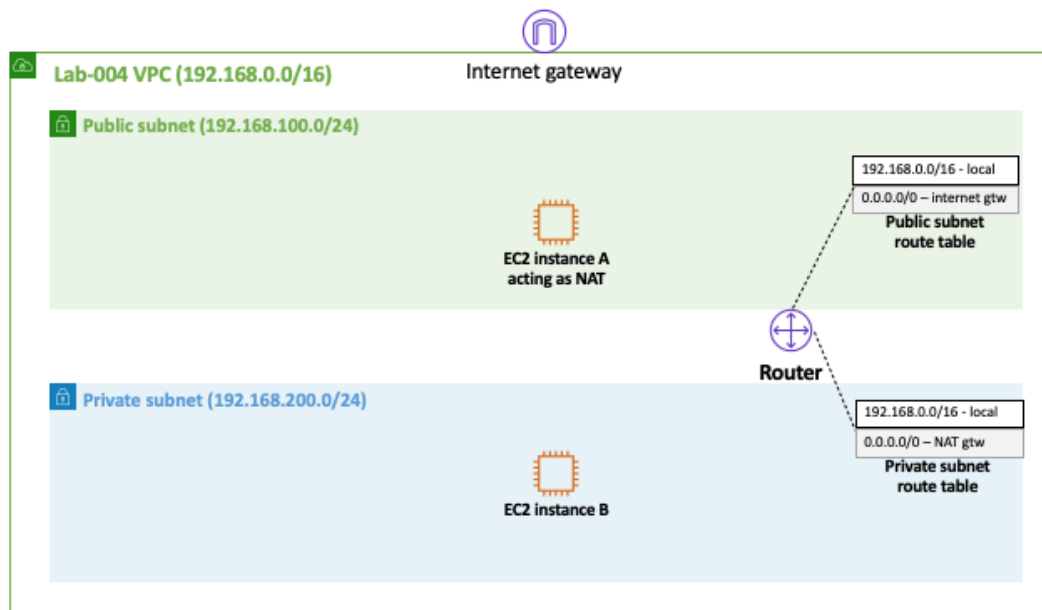
Lab-004

A Single EC2 Instance in a Private Subnet + Bastion Host + Nat Instance

Goal

This lab is similar to [lab-003](#) but instead of using a *Nat gateway* you are asked to use a [Nat instance](#).

Architecture Diagram



view

As mentioned before, this lab is very similar to [lab-003](#) except that you will create a *Nat instance* instead of a *Nat gateway*. To do just that, when you create the EC2 instance in the public subnet (the one labeled B), make sure to choose an AMI that comes with pre-installed NAT software.

Step 1 - Run Lab-002

Repeat all of the steps described in [lab-002](#), making sure that any references to lab-002 are replaced by lab-004. The security group for the instance in the private subnet (the one labeled B) must have an *inbound rule* that enables not only SSH but also HTTP traffic. For the EC2 instance in the public subnet (the one labeled A), follow the specific instructions described in the next step.

Step 2 - Launch an EC2 Instance with NAT Software

Search the community AMIs for one with NAT software pre-installed. After the instance is launched, make sure to enable *port forwarding* by disabling *source/destination check*.

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 1: Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Search by Systems Manager parameter

Quick Start (0) My AMIs (0) AWS Marketplace (23) **Community AMIs (439)** Operating system

Search: nat

1 to 50 of 439 AMIs

Image ID	Image Name	Architecture	Root Device Type	Virtualization Type	ENA Enabled	Actions
amzn-ami-vpc-nat-hvm-2016.03.3.x86_64-ebs - ami-004b0f60	Amazon Linux AMI 2016.03.3 x86_64 VPC NAT HVM EBS	x86_64	ebs	hvm	Yes	Select
amzn-ami-vpc-nat-2018.03.0.20191219.0-x86_64-ebs - ami-01cd9be80fcd58eb1	Amazon Linux AMI 2018.03.0.20191219.0 x86_64 VPC HVM ebs	x86_64	ebs	hvm	Yes	Select

Launch Instance Connect Actions

Filter by tags and attributes or search

Name	Instance ID	Availability Zone	Instance State	Status Checks
instance A	i-0d879afdcf7a76b01	us-east-1c	running	2/2 checks ...

Connect
Get Windows Password
Create Template From Instance
Launch More Like This
Instance State
Instance Settings
Image
Networking
Cloudwatch Monitoring

Change Security Groups
Attach Network Interface
Detach Network Interface
Disassociate Elastic IP Address
Change Source/Dest. Check
Manage IP Addresses

Enable Source/Destination Check

Are you sure that you would like to disable Source/Destination Check for the instance with the following details:

Instance: i-0d879afdcf7a76b01 (instance A)
Network Interface: eni-0adeba5132bcecb6c
Status: Enabled

Cancel **Yes, Disable**

Step 3 - Create a Route Table

Create a new route table with the *Name tag* private and with a default route to the NAT instance created in step 2.

Step 4 - Associate the Route Table to the Private Subnet

Associate the newly created route table to the private subnet.

Test and Validation

To validate this lab you need to access the EC2 instance in the private subnet (labeled as B) and from there try to access the internet, for example:

```
curl www.google.com
```